



Department of Computer Science and Engineering
Jashore University of Science and Technology
Semester Final Examination-2022
1st Year 2nd Semester, Session: 2020-21

Course Code: CSE 1201
Course Title: Data Structure

Time: 3.00 hours
Marks: 72

[N.B. Answer any 06 sets questions from following 08 sets questions.
The figures in the right margin indicate full marks.]

- ✓ 1. a) How can you compute the address of an element in a two-dimensional array? Explain with example(s). (04)
b) Suppose a five dimensional array A is declared using $A(2:8, -4:1, 6:10, 3:7, -2:6)$. Consider the element $A[3, -1, 8, 4, 2]$ in A. Find the effective indices E_1, E_2, E_3, E_4, E_5 and the address of the element in row-major order and column-major order, assuming $\text{Base}(A) = 400$ and there are $w = 4$ words per memory location. (05)
c) Why do we need postfix and prefix notation over infix notation? Translate the infix notation $A * (B + D) / E - F * (G + H / K)$ to its equivalent postfix notation using stack. (03)

- ✓ 2. a) Write the way for minimize overflow in stack with example(s). (03)
b) Consider the priority queue in the following figure which is maintained as a one way list. (06)

	INFO	PRN	LINK
	1 B	2	6
AVAIL →	2		7
	3 D	4	4
	4 E	4	9
START →	5 A	1	1
	6 C	2	3
	7		10
	8 G	5	0
	9 F	4	8
	10		11
	11		12
	12		0

- i) Describe the structure after (X, 2), (Y, 3), (Z, 2) and (W, 1) are added to the queue.
ii) Describe the structure if, after the preceding insertions, 3 elements are deleted.
c) Find the average case time complexity of linear search algorithm that executed on an array contains n elements. (03)

- ✓ 3. a) Consider the linear arrays $A(5:50)$ and $B(1:18)$. Now (04)
i) Find the number of elements in each array.
ii) Suppose $\text{Base}(A) = 300$ and $w = 4$ words per memory cell. Find the address of $A[35]$ and $A[55]$.
b) Consider a hash table of size 7 with hash function $h(k) = k \bmod 7$. Draw the table that results after inserting in the given order, the values, 19, 26, 13, 48, 17 for each of the scenarios below: (04)
i) When collisions are handled by separate chaining.
ii) When collisions are handled by linear probing.
iii) When collisions are handled by double hashing.
Using a second hash function $h'(k) = 5 - (k \bmod 5)$.
c) Write some advantages and disadvantages of using linked storage for storing strings. (04)

- ✓ 4. a) Compress the sentence "Happy hip hop" using Huffman coding scheme and also show the effectiveness of the compression. (06)

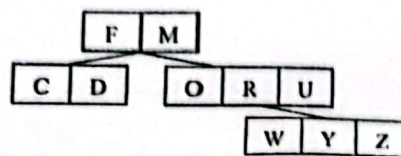
- b) Suppose queue is stored in a circular array with $N=10$ memory cells. (06)
- Find the number of elements in a queue when $FRONT=2$, $REAR=6$.
 - When will the array be filled?

- ✓ 5. a) What is deque? Consider the following deque of characters where deque is a circular array which is allocated 8 memory cells: $LEFT=7$, $RIGHT=2$. (05)

yy	zz	-	-	-	-	ww	xx
----	----	---	---	---	---	----	----

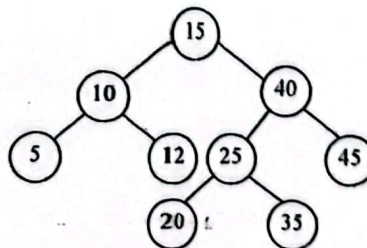
Describe the deque which the following operations take place:

- FF is added to the right of the deque.
 - KK and LL are added to the left of the deque.
 - One letter on the left is deleted.
- b) The following figure is represented as a 4-way search tree. Draw the final tree after deletion of U and M. (03)



- c) What are the moves allowed by the solution of "Tower of Hanoi" problem when $n=4$ disks. (04)
6. a) Give any two representations of graph. (04)
- b) Explain Huffman's algorithm. (04)
- c) Illustrate the working of Heap sort algorithm on the following input: 35, 15, 10, 9, 60. (04)

- ✓ 7. a) Consider the following AVL tree: (04)



- Insert the key 30 into the tree and rebalance if needed. Draw the final and all intermediate trees that you need.
 - Remove the key 40 from the original tree and rebalance if needed. Draw the final tree and all intermediate trees that you need.
- b) Prove that the height of a complete binary tree with N node is about $\log_2 N$. (03)
- c) Suppose a data set S contains n elements. Compare the time complexity $T[1]$ of Linear search algorithm with the time complexity $T[2]$ of the binary search algorithm when (i) $n=1000$ and (ii) $n=10000$. (05)

- ✓ 8. a) Define sorting? Mention types of sorting. (02)
- b) Illustrate the insertion sort algorithm and bubble sort algorithm on input [30, 20, 10, 60, 70, 40]. (05)
- c) Write an algorithm for sorting data in order using selection sort and calculate the value of complexity of this algorithm. Write some drawbacks of this algorithm. (05)



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Semester Final Examination-2022

B.Sc. (Engg.) 1st Year 2nd Semester, Session: 2020-21

Course Code: EEE 1201

Time: 3.00 hours

Course Title: Electronic Devices and Circuits

Marks: 72

[N.B. Answer any 6 set questions from the following 8 set questions. The figures in the margin indicate full marks.]

- ✓ 1. (a) Compare conductor, insulators and semiconductor using band theory

3

(b)

Determine the dc resistance levels for the diode (characteristics curve) of Fig. 1 at

- I. $I_D = 2 \text{ mA}$ (low level)
- II. $I_D = 20 \text{ mA}$ (high level)
- III. $V_D = -10 \text{ V}$ (reverse-biased)

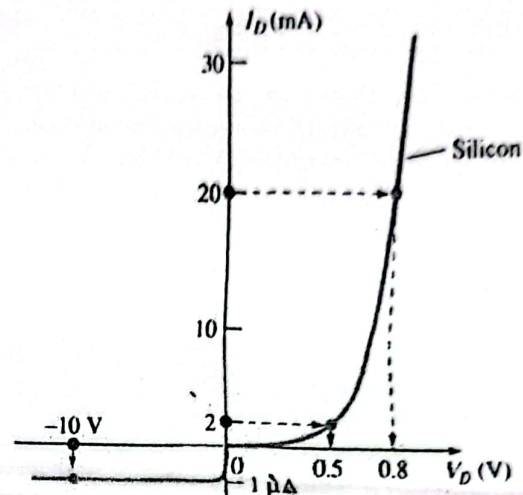


Fig: 1

- (c) Prove that - "The higher the current through a diode, the lower is the dc resistance level."

3

- ✓ 2. (a) Derive the equation of energy stored in a parallel plate capacitor

4

- (b) Calculate the output voltage of a noninverting amplifier (as in Fig. 2) for values of $V_1 = 2\text{V}$, $R_f = 500\text{k}\Omega$, and $R_1 = 100 \text{ k}\Omega$

4

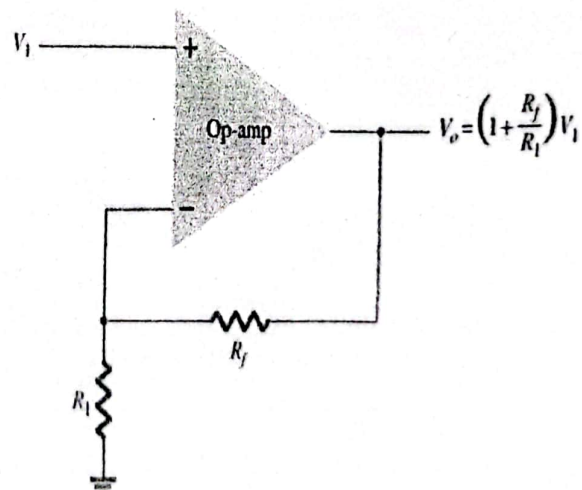


Fig: 2

- (c) For the Zener diode network of Fig. 3, determine V_L , V_R , I_Z , and P_Z

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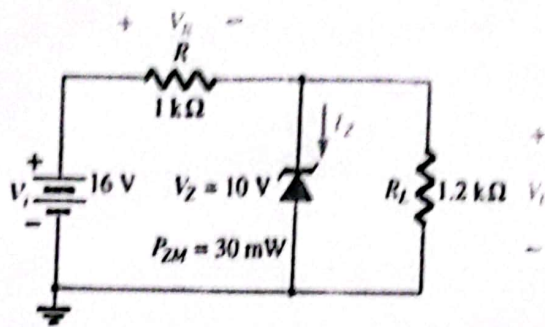


Fig: 3

3. (a) Explain Early effect in BJT
 (b) Draw the ac equivalent circuit, the hybrid equivalent model and the approximate hybrid equivalent model of the following BJT circuit in Fig: 4

2
 2+4+4

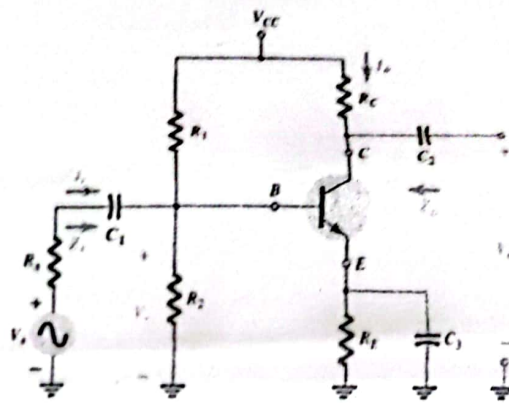


Fig: 4

- ✓ (a) Determine the currents I_1 , I_2 , and I_{D2} for the network of Fig 5

6

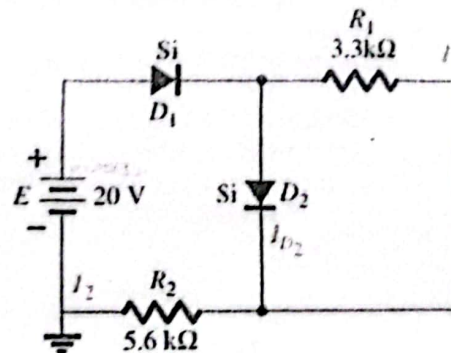


Fig: 5

- (b) Determine the output waveform for the network of Fig. 6 and calculate the output dc level and the required PIV of each diode.

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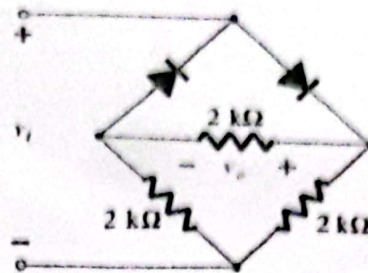
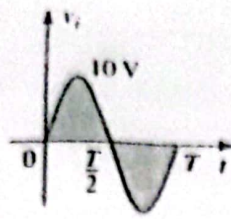


Fig: 6

✓ 5. (a) "Minority charge carrier do not have any obstacle to move through depletion region" - justify 3

(b) Explain the following Fig 7 in accordance with (intrinsic carrier density and relative mobility factor). Propose ways to maximize voltage operating region of diodes. 4

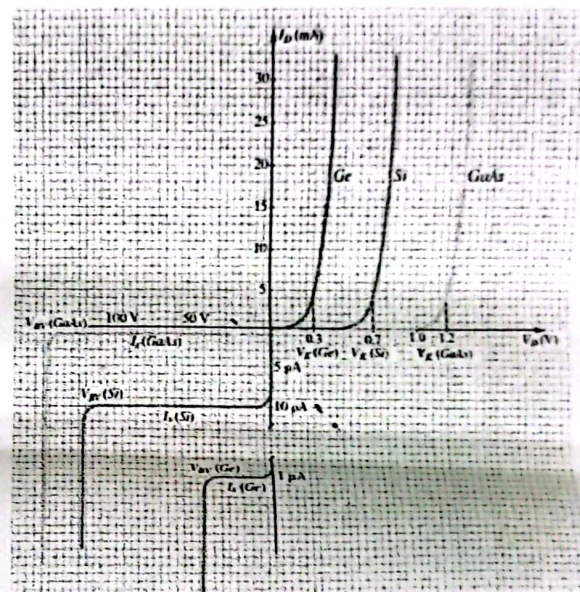


Fig: 7

(c) At a temperature of 27°C determine the thermal voltage V_T 3

(d) "Atomic theory does not work for solids"- Explain 2

✓ 6. (a) Sketch the transfer curve for a p-channel device with $I_{DSS} = 4 \text{ mA}$ and $V_p = 3 \text{ V}$ 3

(b) Explain the following diagram in Fig 8. All the symbols contain traditional interpretation. 3

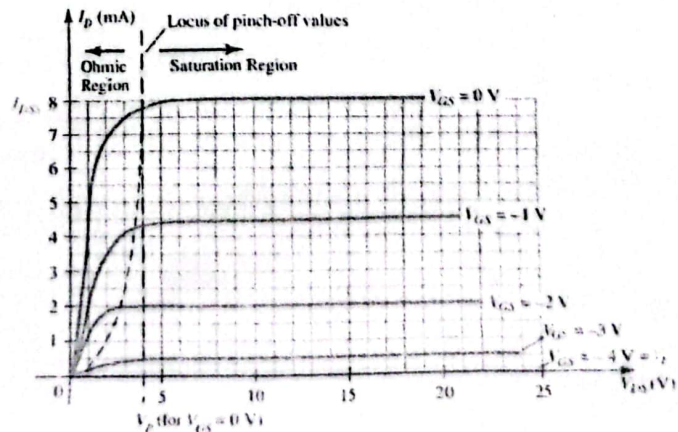


Fig: 8

- (c) Determine the following for the network of Fig. 9 :

I	V_{GSQ}
II	I_{DQ}
III	V_{DS}
IV	V_{GS}
V	V_{DS}
VI	V_S

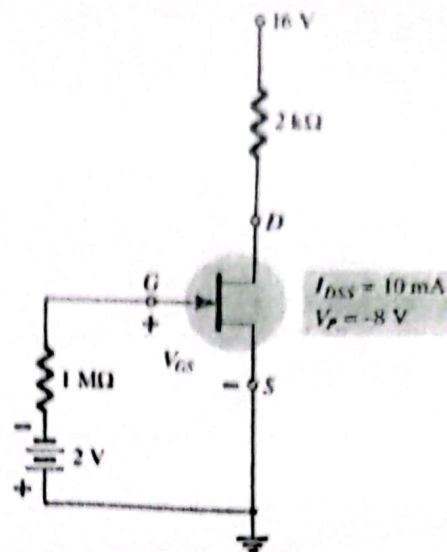


Fig: 9

- (a) Draw the basic diagram and ideal characteristics of *operational amplifiers*? 4
- (b) Derive the expression for *voltage gain non-inverting op-amp*. 4
- (c) Calculate the dc voltages and currents in the circuit of the following Figure. 4

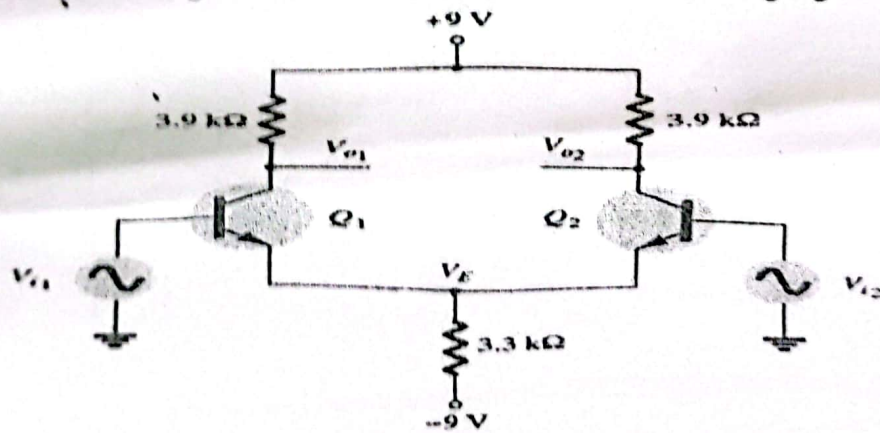


Figure 10

8. (a) Define Capacitor and Capacitance. Briefly explain about *Parallel plate capacitor* and *cylindrical capacitor* with appropriate illustration. 4
- (b) What is the percentage change in capacitance of a parallel plate capacitor when a dielectric slab of dielectric constant 3 is inserted between its plates? 3
- (c) Briefly explain about *voltage-controlled oscillator* with appropriate illustration. 5



Department of Computer Science and Engineering

Jashore University of Science and Technology

Semester Final Examination-2022

1st Year 2nd Semester, Session: 2020-21

Course Code: MATH 1201

Course Title: Differential Equations and Geometry

Time: 3.00 hours

Marks: 72

[N.B. Answer any 06 sets questions from following 08 sets questions. The figures in the right margin indicate full marks.]

- ✓ 1. a. Define with examples: 4
- i) Linear and nonlinear differential equations
 - ii) Order and degree of a differential equations
- b. Eliminate arbitrary constant from $\sqrt{(1-x^2)} + \sqrt{(1-y^2)} = a(x-y)$. 4
- c. Find the differential equation of all circles passing through the origin and having their centers on the x-axis. 4
- ✓ 2. a. Solve the differential equation: $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$. 3
- b. Define homogeneous differential equation. solve: $y^2 + x^2 \frac{dy}{dx} = xy \frac{dy}{dx}$. 4
- c. Define linear differential equation. Solve: $x \frac{dy}{dx} + 2y = x^2 \log x$. 5
3. a. Define exact differential equation. Find the necessary and sufficient condition for a differential equation of first degree being exact. 7
- b. Identify the differential equation $(x - 2e^y)dy + (y + x \sin x)dx$ being exact or not? If exact then find its solution. 5
- ✓ 4. a. State and prove the necessary and sufficient conditions for the exactness of the differential equation $M(x, y)dx + N(x, y)dy = 0$. 6
- b. Define orthogonal trajectory. Find the orthogonal trajectory of the family of curves $y = cx^3$. 6
- ✓ 5. a. Find the condition that the general equation of second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ may represent a pair of straight line. 6
- b. Prove that the straight lines represented by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will be equidistant from the origin if $f^4 - g^4 = c(bf^2 - ag^2)$. 6
- ✓ 6. a. Show that the straight line $y = mx \pm a\sqrt{(1+m^2)}$ is always tangents to the circle $x^2 + y^2 = a^2$. 6
- b. Prove that the locus of the poles of tangents to the circle $x^2 + y^2 = a^2$ with respect to the circle $x^2 + y^2 = 2bx$ is the conic $(a^2 - b^2)x + a^2y^2 - 2a^2bx + a^2b^2 = 0$. 6
- ✓ 7. a. Define radical axis of the two circles. Find the equation of the radical axis of the circle where $S_1 - S_2 = 0$. 7
- b. Find the radical axis of the pair of circles $x^2 + y^2 = 144$ and $x^2 + y^2 - 15x + 11y = 0$. 5
8. a. Find the equation of the parabola whose focus is the point $(1, -1)$ and whose directrix is the line $3x - 4y + 1 = 0$. 6
- b. If l and l' are the length of the segments of any focal chord of the parabola $y^2 = 4ax$ then prove that $\frac{1}{l} + \frac{1}{l'} = \frac{1}{a}$. 6

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) Simplify these expressions into simplified SOP form: 12
- a) $W = AA + AB' + A'BB'C$
- b) $Y = (A+B'+C'+D')' + A'B + CD + AC'E' + AC'$

2. a) Draw the truth table of the following Boolean Function. Then draw the Karnaugh Map to find the simplified Boolean Expression. Finally draw the logic diagram of - 12

$$F(A,B,C,D) = \sum(0,1,2,6,9,12) + \sum d(3,4,8,13)$$

- ✓ 3. a) Number system is a writing system of expressing numbers. To design efficient digital systems an engineer must have the knowledge of number system conversion. Also, s/he must be able to represent negative binary numbers using sign and magnitude, 1's complement and 2's complement methods. Now, Answer the following questions. 06

- I. Convert the number to its equivalent form: $(242.625)_{10} = (?)_2$.
- II. Convert the number to its equivalent form: $(AD6D)_{16} = (?)_8$.
- III. Perform the arithmetic operation using 1's complement:

$$\text{iv. } (-87)_{10} + (-23)_{10} = (?)_{10}$$

Assume that, the representation is an 8-bit register.

- b) Determine the value of 'x' if $(211)_x = (152)_8$. 04

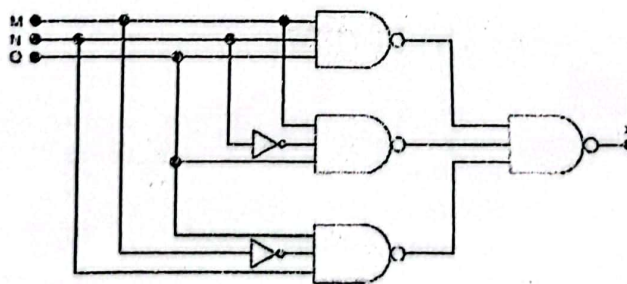
- c) Represent decimal number 8620 in i) BCD and ii) excess-3 code. 02

- ✓ 4. a) Design a 2-to-4 Binary Decoder with proper diagram, truth table and circuit diagram. 6

- b) The decoder is an electronic device that is used to convert a digital signal to an analog signal. It allows a single input line and produces multiple output lines. The decoders are used in many communication projects that are used to communicate between two devices. How a decoder can be used for Automatic Wireless Health Monitoring System in Hospitals for Patients 6

- ✓ 5. a) Design a 4:1 Multiplexer. Also describe Advantages and Disadvantages of Multiplexer. 12

- ✓ 6. a) 05



- I. Simplify the circuit of figure using Boolean algebra.
- II. Change each gate in Problem to a NOR gate, and simplify the circuit using Boolean algebra.

- b) Construct a 16×1 mux using five 4×1 mux only. 04

- c) Prove that $f_1 = m_0 + m_2 + m_5 + m_7 = M_1.M_3.M_4.M_6$. 03

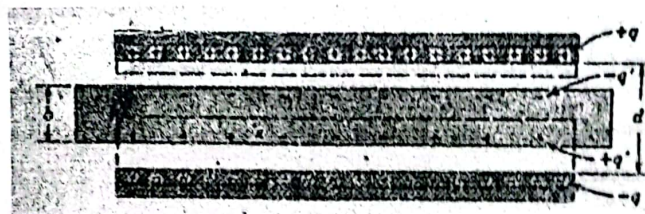
- ✓ 7. a) Differentiate between Latch and Flip Flop. 6

- b) Describe D-Flip Flop. Also show the advantages of using D-Flip Flop instead of S-R Flip Flop 6

- ✓ 8. a) What is shift register? Show the structure of Parallel IN Serial OUT shift register 6

- ✓ 1. a) Define electric potential difference. Find electric potential energy of a system of charges q_1 and q_2 at distance r apart. 1+3
- b) Calculate Electric field (E) from Electric potential (V). 04
- c) Two protons in a nucleus of U^{238} are $6.0 \times 10^{-15} \text{ m}$ apart. What is their mutual electric potential energy? 02
- d) What is the electric potential at the surface of a gold nucleus? The radius is $6.6 \times 10^{-15} \text{ m}$ and the atomic number $Z = 79$. 02
- ✓ 2. a) Prove that, the total energy of a particular executing simple harmonic motion is proportional to the square of the amplitude of the motion. 04
- b) A block whose mass is 680 gm is at rest on a table top and is fastened to an anchored horizontal spring. The spring constant of the spring is 65 N/m . There is negligible friction between the block and the table top. The block is pulled a distance $x = 11 \text{ cm}$ from its equilibrium position at $x = 0$ and released from rest at $t = 0$. 08
- (i) What force does the spring exert on the block just before the block is released?
- (ii) What are the angular frequency, the frequency, and the period of the resulting oscillation?
- (iii) What is the amplitude of the oscillation?
- (iv) What are the maximum velocity and acceleration of the oscillating block?
- (v) What is the mechanical energy of the oscillating system?
- (vi) What are the potential and kinetic energies of the oscillator when $y = dy/dx$?
- ✓ 3. a) Prove the statement, the average translation kinetic energy of molecules in random motion in an ideal gas is directly proportional to absolute temperature of gas. 04
- b) Define Ideal gas. Drive the equation when heat is transferred at a steady rate through a composite slab in conduction with proper figure. 1+3
- c) Determine the volume of 1.00 mol of any gas, assuming it behaves like an ideal gas, at STP. 02
- d) What is the average translation kinetic energy of molecules in an ideal gas at 47° C ? 02
- ✓ 4. a) Discuss the Huygens' principle in interference of light. 03
- b) Light from a sodium vapour lamp ($\lambda = 589 \text{ nm}$) forms an interference pattern on a screen 0.8 m from a pair of slits. The bright fringes in the pattern are 0.35 cm apart. What is the slit separation? 03
- c) Write the conditions for interference of light? 03
- d) In a Young's double slit experiment, the separation between the sources is 0.18 mm . The fringes are observed on a screen 90 cm away. If with certain monochromatic source of light, the third bright fringe is situated at a distance of 8.1 mm from the central bright fringe. Find the wavelength of light. 03

5. a) Why and how acoustically dead rooms are created? Write down requirements to improve the acoustics of an auditorium. 06
- b) Due to efficiency considerations related to its bow wake, the supersonic transport aircraft must maintain a cruising speed that is a constant ratio to the speed of sound (a constant Mach number). If the aircraft flies from warm air into colder air, should it increase or decrease its speed? Explain your answer. 01
- c) Pipe A is open at both ends and has length $L_A = 0.343$ m. We want to place it near three other pipes in which standing waves have been set up, so that the sound can set up a standing wave in pipe A. Those other three pipes are each closed at one end and have lengths $L_B = 0.500L_A$, $L_C = 0.250L_A$, and $L_D = 2.00L_A$. For each of these three pipes, which of their harmonics can excite a harmonic in pipe A? 01
6. a) Define Gauss's law? Derive Coulomb's from Gauss's law. 1+3
- b) A hypothetical Cylinder of radius R immersed in a uniform electric field E, the cylinder axis being parallel to the field. What is the Φ_E for this closed surface 04
- c) At one time the positive charge in the atom was thought to be distributed uniformly throughout a sphere with a radius of about 1.0×10^{-10} meters, that is, throughout the entire atom. Calculate the electric field strength at the surface of a platinum atom ($Z = 78$). Neglect the effect of the electrons. 02
- d) An infinite rod of charge in which the linear charge density is λ (that is, the charge per unit length, measured in coul/meter) being constant for all points on the line. Find the expression for E at distance from the line. 02
- ✓ a) Define electric field. Prove that, electric field E exists at any point in space, we can think of that point as the site of stored energy in amount, per unit volume, of $\frac{1}{2} \epsilon_0 E^2$. 1+5
- b) 06



Following figure shows a dielectric slab of thickness b and dielectric constant k placed between the plates of a parallel-plate capacitor of plate area A and separation d . A potential Difference V_0 is applied with no dielectric present. The battery is then disconnected and the dielectric slab inserted. Assume $A = 100 \text{ cm}^2$, $d = 1.0 \text{ cm}$, $b = 0.50 \text{ cm}$, $k = 7.0$, and $V_0 = 100$ volts.

- Calculate the capacitance C_0 before the slab is inserted.
- Calculate the free charge of q .
- Calculate the electric field strength in the gap.
- Calculate the electric field strength in dielectric.
- Calculate the potential difference between the plates.
- Calculate the capacitance with the slab in place

- ✓ 8. a) State the characteristics of a standing wave and the conditions of forming standing wave. 04
- b) Briefly describe the phenomenon of Interference of sound waves. 05
- c) Deduce the resultant displacement that happens when two simple harmonic waves interfere with each other and also discuss the conditions that show how phase 03